STPM 2016 PHYSICS 960/2 (U2)

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg, $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1}, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9$ N m^2 C^{-2}, $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}.

Section A [15 marks]

1. Two point charges, $-Q$ and $-2Q$, repel each other with a force of magnitude F_0. Extra charges of $-Q$ are then added to each of the point charges and the distance of separation between the charges is doubled. The new force, in terms of F_0, that exists between the charges is

A $\frac{1}{4}F_0$ B $\frac{1}{2}F_0$ C $\frac{3}{4}F_0$ D $\frac{3}{2}F_0$

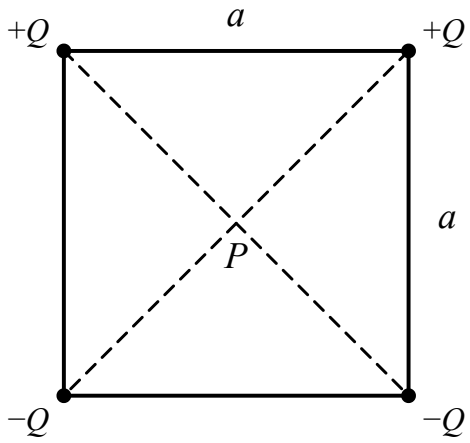
Answer: C

$$F_0 = \frac{k(Q)(2Q)}{r^2} = \frac{2kQ^2}{r^2}$$

$$\text{New: } \frac{k(2Q)(3Q)}{(2r)^2} = \frac{1.5kQ^2}{r^2} = \frac{3}{4}F_0$$

Explanation: The charges become $-2Q$ and $-3Q$.

2. Two point charges $+Q$ and another two point charges $-Q$ which are located at the four vertices of a square of dimension a are shown in the diagram below.



What is the electric field strength and electric potential at P?

	Electric field strength	*Electric potential*
A	0	0
B	0	$\frac{\sqrt{2}Q}{\pi\varepsilon_0 a}$
C	$\frac{\sqrt{2}Q}{\pi\varepsilon_0 a^2}$	0
D	$\frac{\sqrt{2}Q}{\pi\varepsilon_0 a^2}$	$\frac{\sqrt{2}Q}{\pi\varepsilon_0 a}$

Answer: C

Potential: two $+Q$ and two $-Q$ at equal distances, so $V = 0$.

Each charge is $\frac{a}{\sqrt{2}}$ from P: field of each = $\frac{kQ}{a^2/2} = \frac{2kQ}{a^2}$. Each diagonal pair ($+Q$ and the opposite $-Q$) gives $\frac{4kQ}{a^2}$; the two diagonals are perpendicular

$$E = \sqrt{2} \times \frac{4kQ}{a^2} = \frac{\sqrt{2}Q}{\pi\varepsilon_0 a^2}$$

Explanation: The fields of opposite charges on a diagonal point the same way at the centre.

3. The plates of an isolated charged capacitor are pulled further apart. Which statement is **not** true of the capacitor?

A The capacitance of the capacitor decreases.
B The energy stored in the capacitor increases.
C The charge on the capacitor remains constant.
D The potential difference across the capacitor remains constant.

Answer: D

Isolated: Q is constant. Pulling the plates apart decreases C, so $V = Q/C$ increases. Statement D (p.d. constant) is not true.

Explanation: The energy $Q^2/2C$ increases: work is done pulling the plates apart.

4. The potential difference across a capacitor of 2200 μF decreases to 25% from its initial value in time T, when the capacitor is discharged through a resistor of 25 kΩ. What is the value of T?

A 14 s B 16 s C 41 s D 76 s

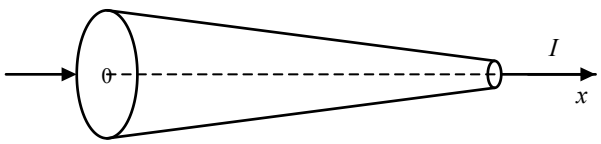
Answer: D

$$V = V_0 e^{-t/RC}: 0.25 = e^{-T/RC} \Longrightarrow T = RC \ln 4$$

$$T = (25 \times 10^3)(2200 \times 10^{-6})(1.386) = 76 \text{ s}$$

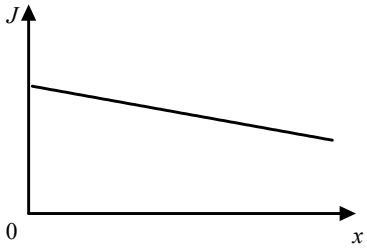
Explanation: $RC = 55$ s; two half-lives (2 × 38 s) also give 25%.

5. A conductor of a cross-sectional area which decreases with a distance x is shown in the diagram below.

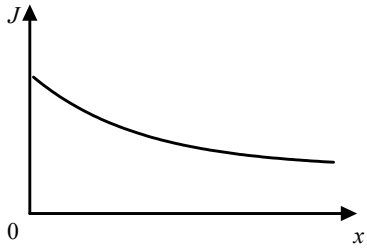


When a constant current I flows through the conductor, which graph shows the variation of current density J with the distance x?

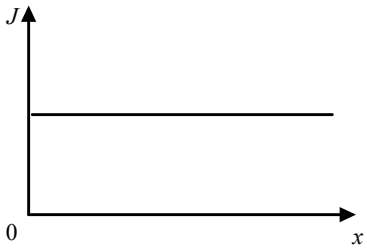
A



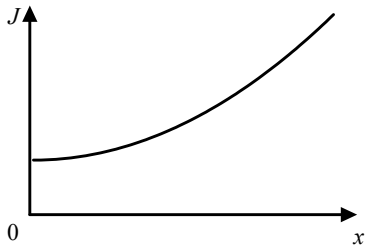
B



C



D



Answer: D

$J = \frac{I}{A}$: A decreases (as the square of the radius), so J increases more and more steeply (D)

Explanation: The current is the same at every cross-section.

6. When an electric field is applied to a metal conductor, the free electrons drift with a velocity v. If the mean time between collisions is t, the mass of electron is m, the charge of electron is e and the electric field strength is E, the correct expression for v is

A $\frac{eEt}{2m}$ B $\frac{eEt}{m}$ C $\frac{3eEt}{4m}$ D $\frac{2eEt}{m}$

Answer: B

Acceleration $\frac{eE}{m}$ for time t between collisions: $v = \frac{eEt}{m}$

Explanation: This is the drift velocity in the simple model.

7. A potential divider is used to

A produce a constant voltage from a source of current
B increase the voltage supplied by a source of current
C produce half of the voltage supplied by a source of current
D produce a fraction of the voltage supplied by a source of current

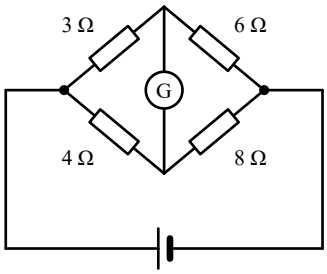
Answer: D

A potential divider gives a chosen fraction of the supply voltage (D).

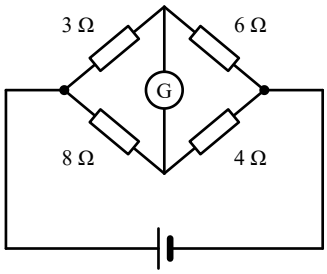
Explanation: $V_{out} = V_{in} \times R_2/(R_1 + R_2)$.

8. Which galvanometer in the circuits below shows zero deflection?

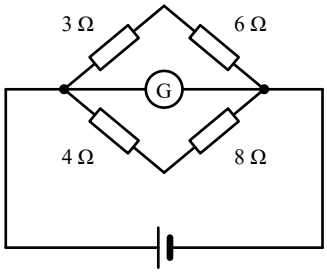
A



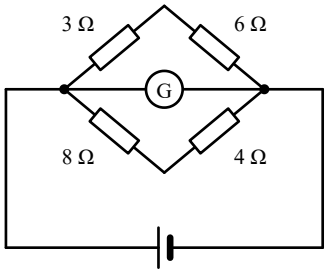
B



C



D

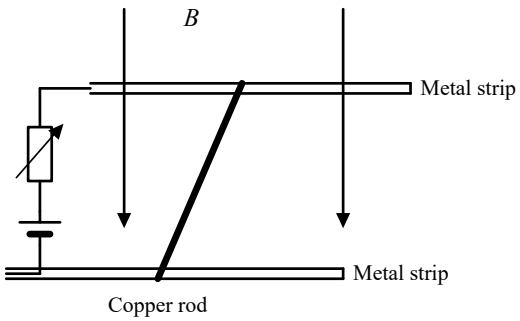


Answer: A

A: $\frac{3}{6} = \frac{4}{8}$: the bridge is balanced, so G shows zero deflection

Explanation: In balance, the ratios of resistances on the two sides of the galvanometer are equal.

9. A copper rod of length 10 cm is located on two fixed horizontal metal strips which are connected to a battery as shown in the diagram below.



A magnetic field B of flux density 2.0 T is applied vertically downwards. If the limiting static friction between the copper rod and the metal strips is 0.50 N, what is the minimum current that will move the copper rod?

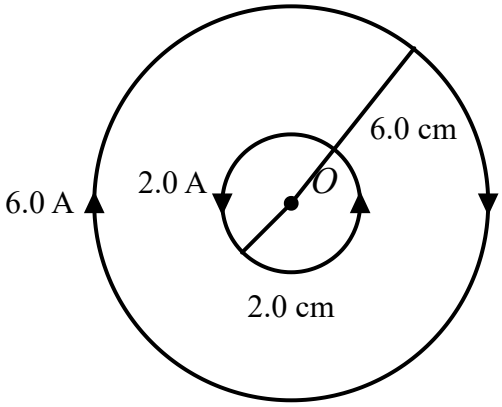
A 5.0 A B 2.5 A C 0.50 A D 0.25 A

Answer: B

$$BIL = 0.50 \text{ N} \Longrightarrow I = \frac{0.50}{(2.0)(0.10)} = 2.5 \text{ A}$$

Explanation: The magnetic force must just exceed the limiting friction.

10. Two concentric coils of radii 2.0 cm and 6.0 cm respectively are shown in the diagram below.



Each coil has 100 turns. If the small coil and the bigger coil carry a current of 2.0 A and 6.0 A respectively in the direction as shown in the diagram, the resultant magnetic flux density at centre O is

A 0 B 2.0×10^{-3} T C 6.3×10^{-3} T D 1.3×10^{-2} T

Answer: A

$B = \frac{\mu_0 NI}{2r}$: small coil $\frac{(4\pi \times 10^{-7})(100)(2.0)}{2(0.020)} = 6.3 \times 10^{-3}$ T; large coil $\frac{(4\pi \times 10^{-7})(100)(6.0)}{2(0.060)} = 6.3 \times 10^{-3}$ T

The currents circulate in opposite directions, so the fields cancel: $B = 0$.

Explanation: B at the centre is proportional to I/r, which is the same for both coils.

11. Which statement is **not** true about Hall effect?

A Hall effect can be used to determine the sign of charge carriers.
B Hall voltage of typical order of microvolt for metal is produced.
C Hall voltage is more significant when a thicker specimen is used.
D Hall effect can be used to determine the number of charge carriers per unit volume.

Answer: C

$V_H = \frac{BI}{nqt}$: a thicker specimen gives a smaller Hall voltage, so C is not true

Explanation: Thin specimens are used for Hall probes.

12. Which statement is **not** true about magnetic flux?

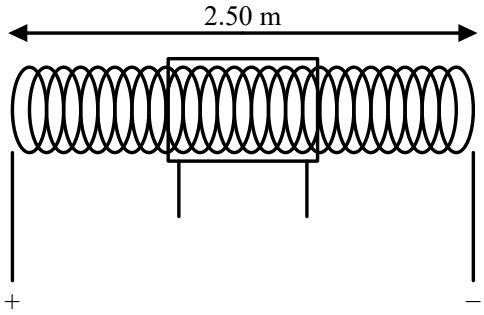
A It is independent of surface area.
B It is zero if the magnetic field is parallel to the surface.
C It represents the total magnetic lines of force through a surface.
D It is the product of the normal component of magnetic flux density with the surface area.

Answer: A

$\Phi = BA\cos\theta$: the flux depends on the area, so A is not true

Explanation: Flux is zero when **B** is parallel to the surface.

13. A coil of 200 turns wound over the middle part of a long solenoid of 1200 turns is shown in the diagram below.



The cross-sectional area of the coil and the solenoid is 1.25×10^{-4} m^2. If the length of the solenoid is 2.50 m, the mutual inductance between the coil and the solenoid is

A 2.51×10^{-6} H B 1.51×10^{-5} H C 9.05×10^{-5} H D 9.43×10^{-5} H

Answer: B

$$M = \frac{\mu_0 N_1 N_2 A}{l} = \frac{(4\pi \times 10^{-7})(1200)(200)(1.25 \times 10^{-4})}{2.50} = 1.51 \times 10^{-5} \text{ H}$$

Explanation: l is the length of the long solenoid.

14. An electric oven is labelled '240 V, 1200 W'. If the values are taken to be the root mean square, the peak power of the oven is

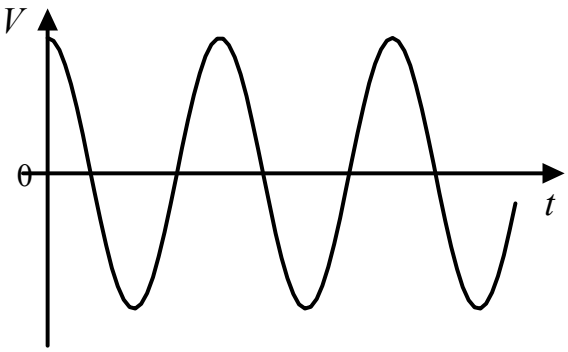
A 600 W B 1200 W C 2400 W D 4800 W

Answer: C

$$\text{Peak power} = V_0 I_0 = (\sqrt{2}V_{\text{rms}})(\sqrt{2}I_{\text{rms}}) = 2 \times 1200 = 2400 \text{ W}$$

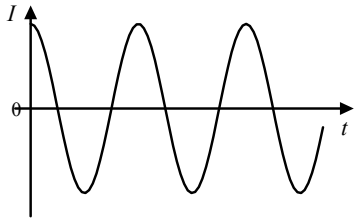
Explanation: The mean power is half the peak power for a resistive load.

15. The a.c. voltage V across a capacitor is represented by the graph below.

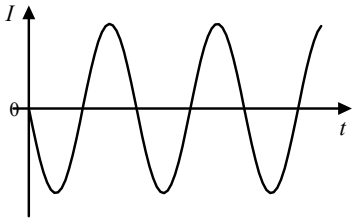


Which graph represents the variation of current I with time t?

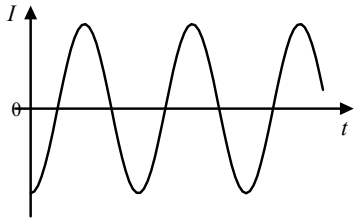
A



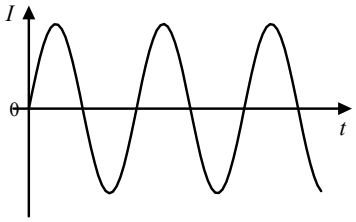
B



C



D



Answer: B

$I = C\frac{dV}{dt}$: V starts at its maximum and is decreasing, so I starts at zero and goes negative (B)

Explanation: In a capacitor the current leads the voltage by 90°.

Section B [15 marks]

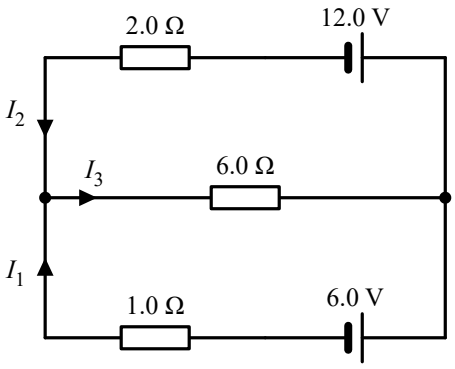
16. (a) State Kirchhoff's laws. [2]

Answer

First law: the algebraic sum of currents at a junction is zero (current in = current out)	1
Second law: around any closed loop, the sum of the e.m.f.s equals the sum of the IR products	1
	2

Explanation: They express conservation of charge and of energy.

(b) A circuit which has three resistors of 2.0 Ω, 6.0 Ω and 1.0 Ω connected to two batteries of 12.0 V and 6.0 V is shown in the diagram below.



(i) Write three equations by using Kirchhoff's laws. [3]

Answer

$I_3 = I_1 + I_2$ 1

Upper loop: $12.0 = 2.0I_2 + 6.0I_3$ 1

Lower loop: $6.0 = 1.0I_1 + 6.0I_3$ 1

3

Explanation: Each loop includes the shared 6.0 Ω resistor.

(ii) Determine the current I_2. [1]

Answer

Solving: $I_3 = 1.2$ A, $I_1 = -1.2$ A, $I_2 = 2.4$ A 1

1

Explanation: I_1 is negative: it flows opposite to the arrow (the 6.0 V cell is being charged).

(iii) Determine the potential difference across the resistor of 6.0 Ω. [2]

Answer

$V = 6.0I_3 = 6.0(1.2)$ 1

$V = 7.2$ V 1

2

Total for question **8**

Explanation: Check with the upper loop: $12.0 - 2.0(2.4) = 7.2$ V.

17. (a) State **two** differences between resistance and impedance. [2]

Answer

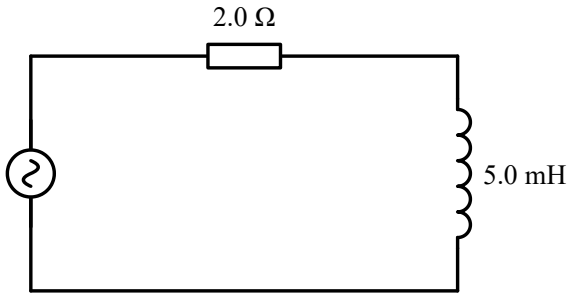
Resistance applies to both d.c. and a.c. and dissipates energy; impedance applies to a.c. and includes reactance, which stores and returns energy 1

Resistance does not depend on frequency; impedance does (through X_L and X_C) 1

2

Explanation: $Z = \sqrt{(R^2 + (X_L - X_C)^2)}$.

(b) A resistor of 2.0 Ω and an inductor of 5.0 mH connected in series to an alternating source are shown in the diagram below.



The current of the alternating source is given by $I = 1.5 \sin (100\pi t)$, where I is current in amperes and t is time in seconds.

(i) Calculate the impedance of the circuit. [3]

Answer

$X_L = \omega L = (100\pi)(5.0 \times 10^{-3}) = 1.57\ \Omega$ 1

$Z = \sqrt{(2.0^2 + 1.57^2)}$ 1

$Z = 2.54\ \Omega$ 1

3

Explanation: $\omega = 100\pi$ rad s^{-1} is read from the current equation.

(ii) Determine the maximum voltage across the circuit. [2]

Answer

$V_0 = I_0 Z = (1.5)(2.54)$ 1

$V_0 = 3.81$ V 1

2

Total for question **7**

Explanation: The peak current is 1.5 A.

Section C [30 marks]

18. (a)

(i) Define *electric potential at a point*. [1]

Answer

The work done per unit positive charge in bringing a small positive test charge from infinity to that point 1

1

Explanation: Unit: volt.

(ii) Two spherical conductors, *X* and *Y*, which are located at a certain distance apart are shown in the diagram below.



X is of radius 0.15 cm and carries a charge of +8.0 μC while *Y* is of radius 0.30 cm and carries a charge of −2.0 μC. Determine the charge which remains on *X* when *X* and *Y* are connected by a wire. [3]

Answer

To be added.

3

(b) An electron is accelerated from rest through a region between electric potential V_1 and electric potential V_2 in vacuum. The final speed of the electron is 2.0×10^6 m s^{-1}.

(i) State the transformation of energy of the electron. [2]

Answer

Electric potential energy of the electron 1

is converted into kinetic energy 1

2

Explanation: $eV = \frac{1}{2}mv^2$.

(ii) Determine the potential difference between V_1 and V_2. [2]

Answer

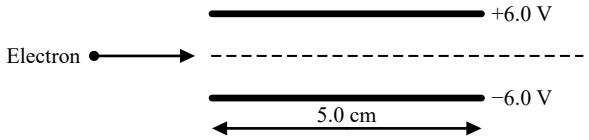
$eV = \frac{1}{2}mv^2 \Rightarrow V = \frac{(9.11 \times 10^{-31})(2.0 \times 10^6)^2}{2(1.60 \times 10^{-19})}$ 1

$V = 11.4$ V 1

2

Explanation: The electron starts from rest.

(c) An electron moves along a line in the middle between a pair of parallel metal plates as shown in the diagram below.



The initial horizontal velocity of the electron is 4.0×10^6 m s^{-1}. The electric potential of the upper and the lower plates are +6.0 V and −6.0 V respectively. The separation between the plates is 1.0 cm.

(i) Calculate the acceleration of the electron between the plates. [4]

Answer

$E = \frac{V}{d} = \frac{6.0 - (-6.0)}{0.010} = 1200$ V m^{-1} 1

$F = eE = (1.60 \times 10^{-19})(1200) = 1.92 \times 10^{-16}$ N 1

$a = \frac{F}{m} = \frac{1.92 \times 10^{-16}}{9.11 \times 10^{-31}}$ 1

$a = 2.11 \times 10^{14}$ m s^{-2}, towards the upper (positive) plate 1

4

Explanation: The p.d. between the plates is 12 V.

(ii) Determine whether the electron can exit from the end of the plates. [3]

Answer

Time between the plates: $t = \dfrac{0.050}{4.0 \times 10^{6}} = 1.25 \times 10^{-8}$ s — 1

Deflection $y = \frac{1}{2}(2.11 \times 10^{14})(1.25 \times 10^{-8})^2 = 1.65 \times 10^{-2}$ m — 1

This is more than 0.50 cm (half the separation), so the electron hits the upper plate and cannot exit — 1

3

Total for question **15**

Explanation: It reaches the plate ($y = 0.50$ cm) after 6.9×10^{-9} s, about 2.8 cm along the 5.0 cm plates.

19. (a) Define *capacitance*. [1]

Answer

The ratio of the charge on either plate to the potential difference between the plates, $C = \dfrac{Q}{V}$ — 1

1

Explanation: Unit: farad (F).

(b) Two square conducting plates of sides 5.50 cm are separated by an air gap of 1.25 cm to form a parallel plate capacitor. The plates are then connected to a supply of 120 V. Determine

(i) the capacitance of the capacitor, [3]

Answer

$A = (0.0550)^2 = 3.03 \times 10^{-3}$ m^2 — 1

$C = \dfrac{\varepsilon_0 A}{d} = \dfrac{(8.85 \times 10^{-12})(3.03 \times 10^{-3})}{1.25 \times 10^{-2}}$ — 1

$C = 2.14 \times 10^{-12}$ F — 1

3

Explanation: Square plates: area = side^2.

(ii) the charge stored in the capacitor. [2]

Answer

$Q = CV = (2.14 \times 10^{-12})(120)$ — 1

$Q = 2.57 \times 10^{-10}$ C — 1

2

Explanation: The supply remains connected.

(c) A parallel plate capacitor is separated by air space of distance 18.5 mm. An electric supply of 12.0 V is then connected to the capacitor and the charge stored in the capacitor is 2.46 μC. A dielectric with dielectric constant of 3.06 is then inserted into the air space between the plates. Determine

(i) the electric field between the plates with air space, [2]

Answer

$E = \frac{V}{d} = \frac{12.0}{18.5 \times 10^{-3}}$ 1

$E = 649 \text{ V m}^{-1}$ 1

2

Explanation: The field between the plates is uniform.

(ii) the capacitance of the plates after the dielectric is inserted, [3]

Answer

$C_0 = \frac{Q}{V} = \frac{2.46 \times 10^{-6}}{12.0} = 2.05 \times 10^{-7} \text{ F}$ 1

$C = \varepsilon_r C_0 = 3.06(2.05 \times 10^{-7})$ 1

$C = 6.27 \times 10^{-7} \text{ F}$ 1

3

Explanation: The dielectric fills the space between the plates.

(iii) the increase in electric potential energy stored in the capacitor before and after the dielectric is inserted. Explain the microscopic mechanism for the increase. [4]

Answer

Before: $U_0 = \frac{1}{2} C_0 V^2 = 1.48 \times 10^{-5}$ J; after: $U = \frac{1}{2} C V^2 = 4.52 \times 10^{-5}$ J 1

Increase = 3.0×10^{-5} J 1

The dielectric molecules are polarised; induced surface charges reduce the field and p.d. between the plates 1

The supply (still connected) drives more charge onto the plates to restore 12.0 V, so more energy is stored 1

4

Total for question **15**

Explanation: With the supply connected, V is constant and $U = \frac{1}{2}CV^2 \propto C$.

20. (a) State **two** laws of electromagnetic induction. [2]

Answer

Faraday: the induced e.m.f. is directly proportional to the rate of change of magnetic flux linkage — 1

Lenz: the induced current flows in a direction that opposes the change producing it — 1

2

Explanation: $\varepsilon = -\mathrm{d}(N\Phi)/\mathrm{d}t$.

(b) A thin circular coil of diameter 30.0 cm and 400 turns is located in a magnetic field such that the plane of the coil is perpendicular to the magnetic field of 40 mT. The magnetic field is then reduced to zero in 2.0 s. Determine

(i) the induced e.m.f. in the coil, [4]

Answer

$A = \pi(0.150)^2 = 7.07 \times 10^{-2}\ \mathrm{m}^2$ — 1

$\varepsilon = NA\dfrac{\Delta B}{\Delta t}$ — 1

$\varepsilon = (400)(7.07 \times 10^{-2})\dfrac{40 \times 10^{-3}}{2.0}$ — 1

$\varepsilon = 0.565\ \mathrm{V}$ — 1

4

Explanation: Use the radius, half of 30.0 cm.

(ii) the magnitude and direction of induced current in the coil if the total resistance of the coil is 0.5 Ω and the magnetic field is directed into the page. [3]

Answer

$I = \dfrac{0.565}{0.5}$ — 1

$I = 1.13\ \mathrm{A}$ — 1

Clockwise, so that its own field is into the page, opposing the decrease (Lenz's law) — 1

3

Explanation: Viewed with the field going into the page.

(c) An air cored long solenoid of length 8.0 cm and diameter 10.0 cm is made of 30.0 turns per cm. A current is flowed and increased to 1.5 A in 0.5 s.

(i) Derive the expression for self inductance of the solenoid. [3]

Answer

$B = \frac{\mu_0 NI}{l}$, flux through each turn $\Phi = BA$ — 1

Flux linkage $N\Phi = \frac{\mu_0 N^2 AI}{l} = LI$ — 1

$L = \frac{\mu_0 N^2 A}{l}$ — 1

3

Explanation: L depends only on the geometry and the core.

(ii) Determine the induced e.m.f. in the solenoid. [2]

Answer

$N = 30.0 \times 8.0 = 240$; $L = \frac{(4\pi \times 10^{-7})(240)^2 \pi (0.050)^2}{0.080} = 7.1 \times 10^{-3}$ H — 1

$\varepsilon = L\frac{\Delta I}{\Delta t} = (7.1 \times 10^{-3})\frac{1.5}{0.5} = 2.1 \times 10^{-2}$ V — 1

2

Explanation: The solenoid is not very long compared with its diameter, so this is only an estimate.

(iii) State the effect on the induced e.m.f. if an iron rod is inserted into the solenoid. [1]

Answer

The induced e.m.f. increases greatly, because the iron core increases the flux (permeability) and so the inductance — 1

1

Total for question **15**

Explanation: $L = \mu N^2 A/l$.